

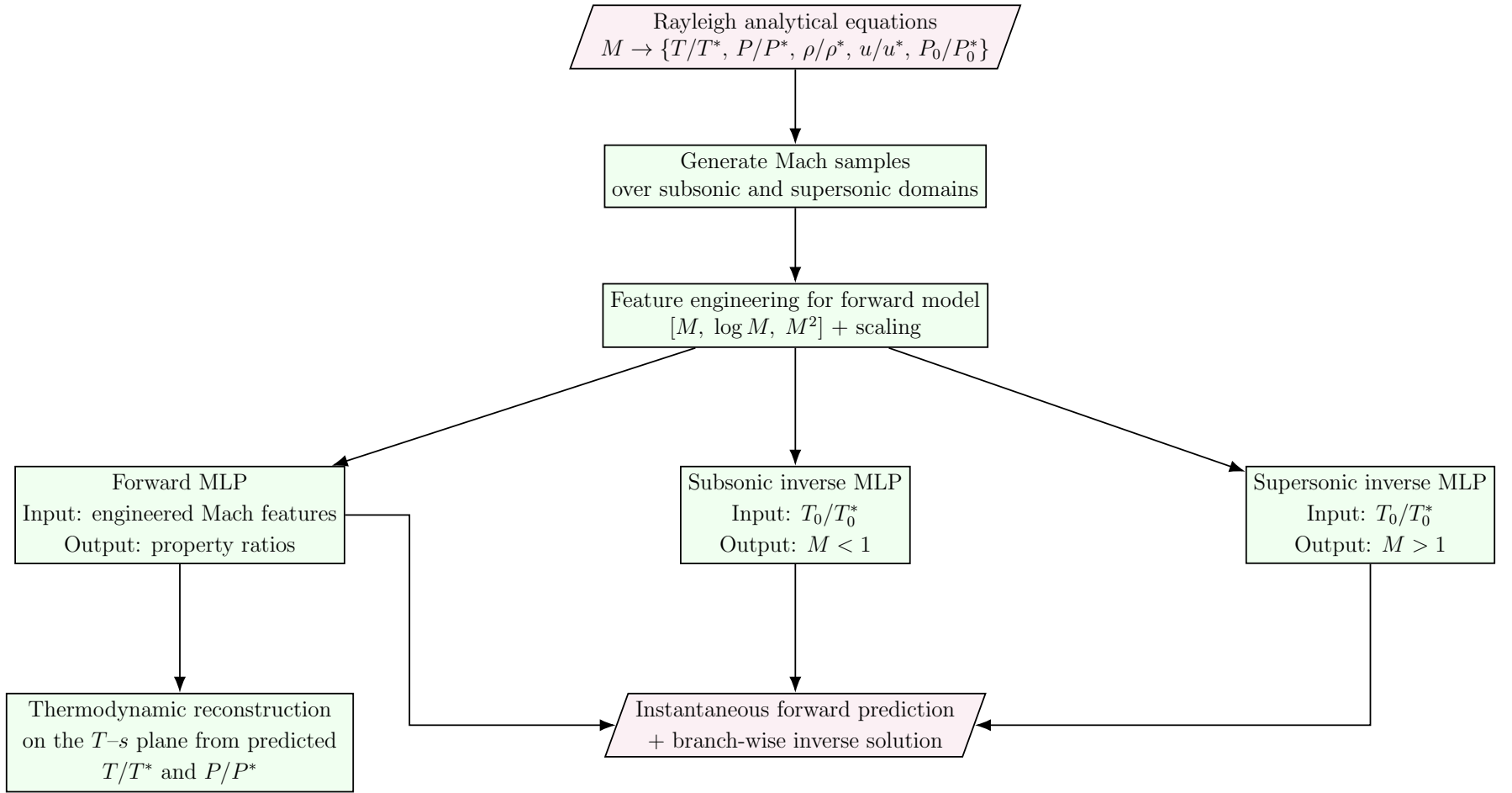


Figure 1: Rayleigh-flow surrogate architecture. The forward model learns the mapping from Mach number to thermodynamic property ratios using engineered Mach-based features, while the inverse problem is resolved through separate subsonic and supersonic expert networks. The forward predictions are additionally used for thermodynamic reconstruction on the T - s plane.

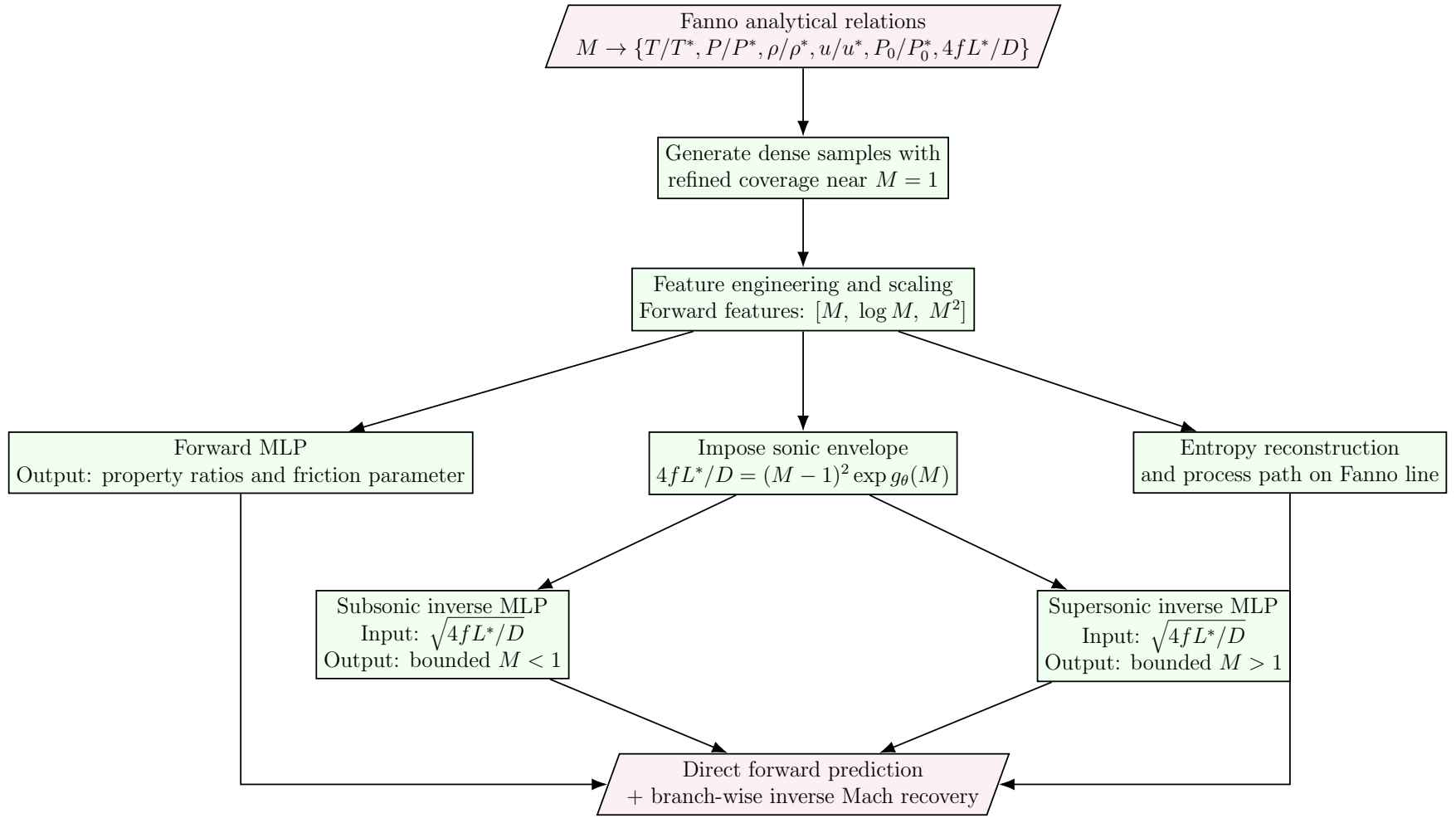


Figure 2: Fanno-flow surrogate architecture. The forward network learns positive thermodynamic ratios and the residual function inside an exact quadratic sonic envelope. The inverse uses the regularized coordinate $\sqrt{4fL^*/D}$ and separate bounded subsonic and supersonic expert networks.

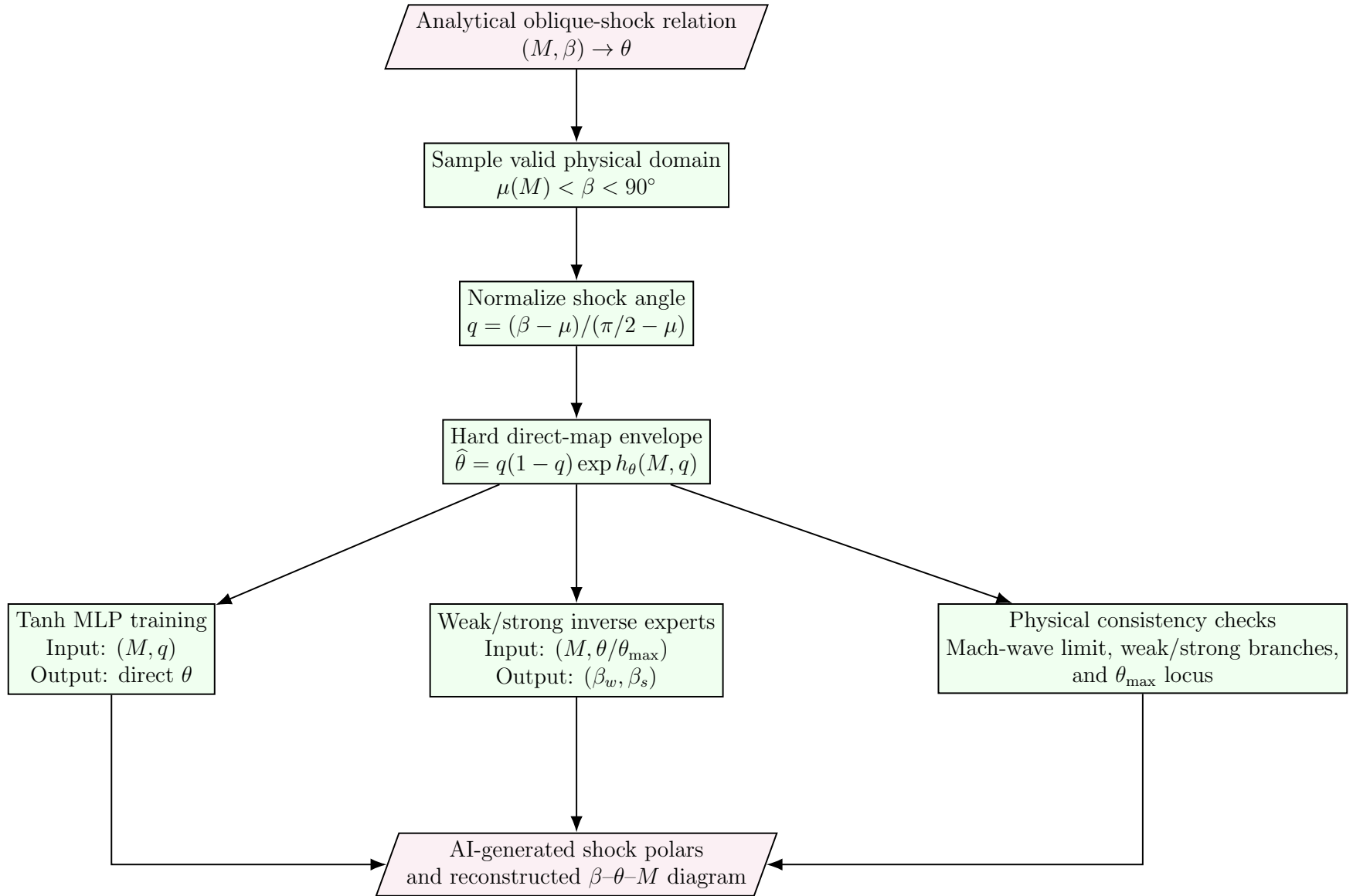


Figure 3: Oblique-shock surrogate workflow. The direct map uses an exact endpoint envelope, while separate bounded inverse outputs represent the weak and strong branches.

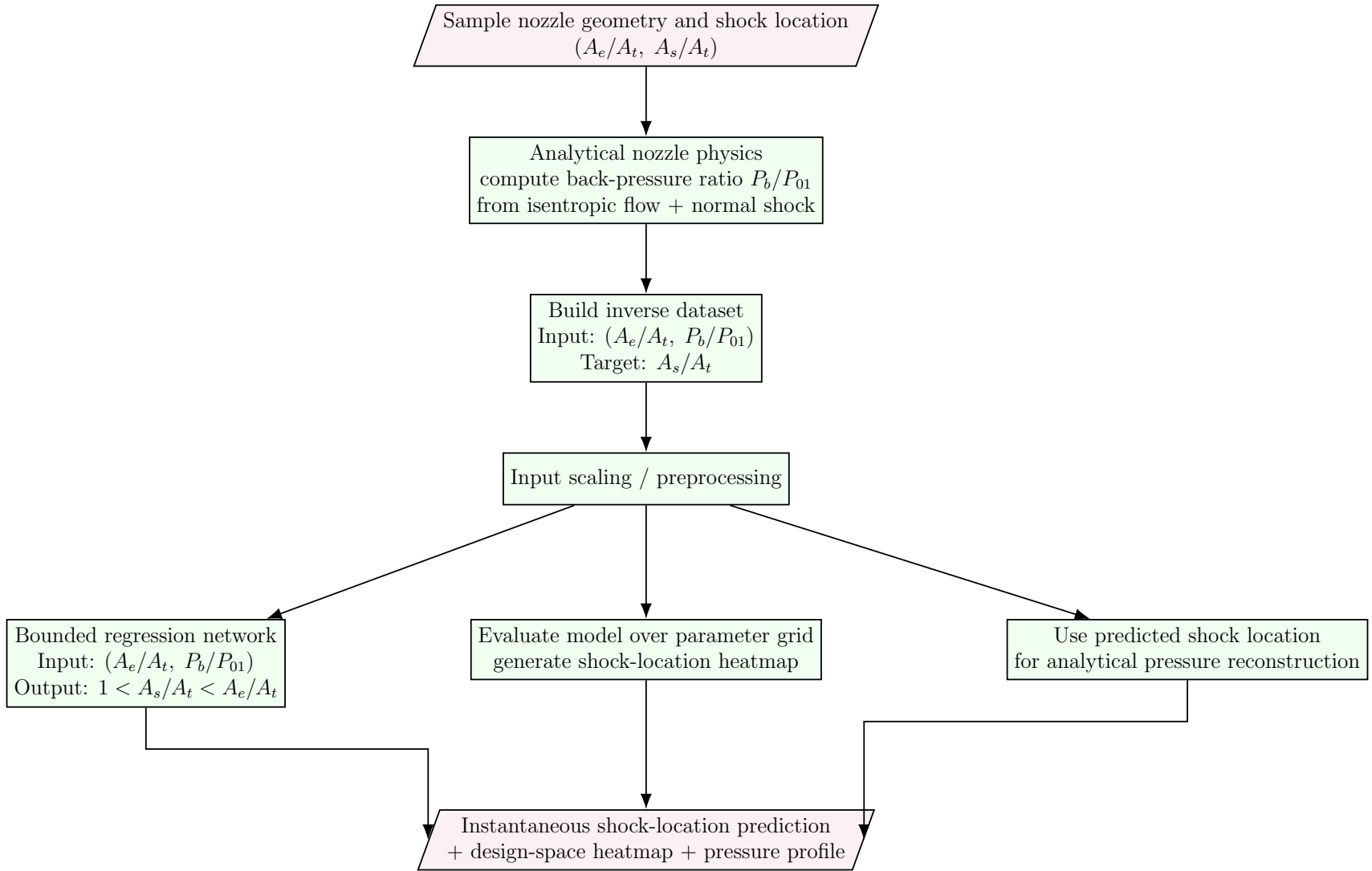


Figure 4: Workflow of the neural-network surrogate for normal-shock location in a convergent-divergent nozzle. Synthetic inverse-design data are generated analytically by sampling nozzle geometries and shock locations, then computing the corresponding back-pressure ratio. A regression network is trained to predict the shock location directly from nozzle geometry and back pressure, and the trained model is subsequently used for real-time design-space heatmaps and analytical pressure-profile reconstruction.

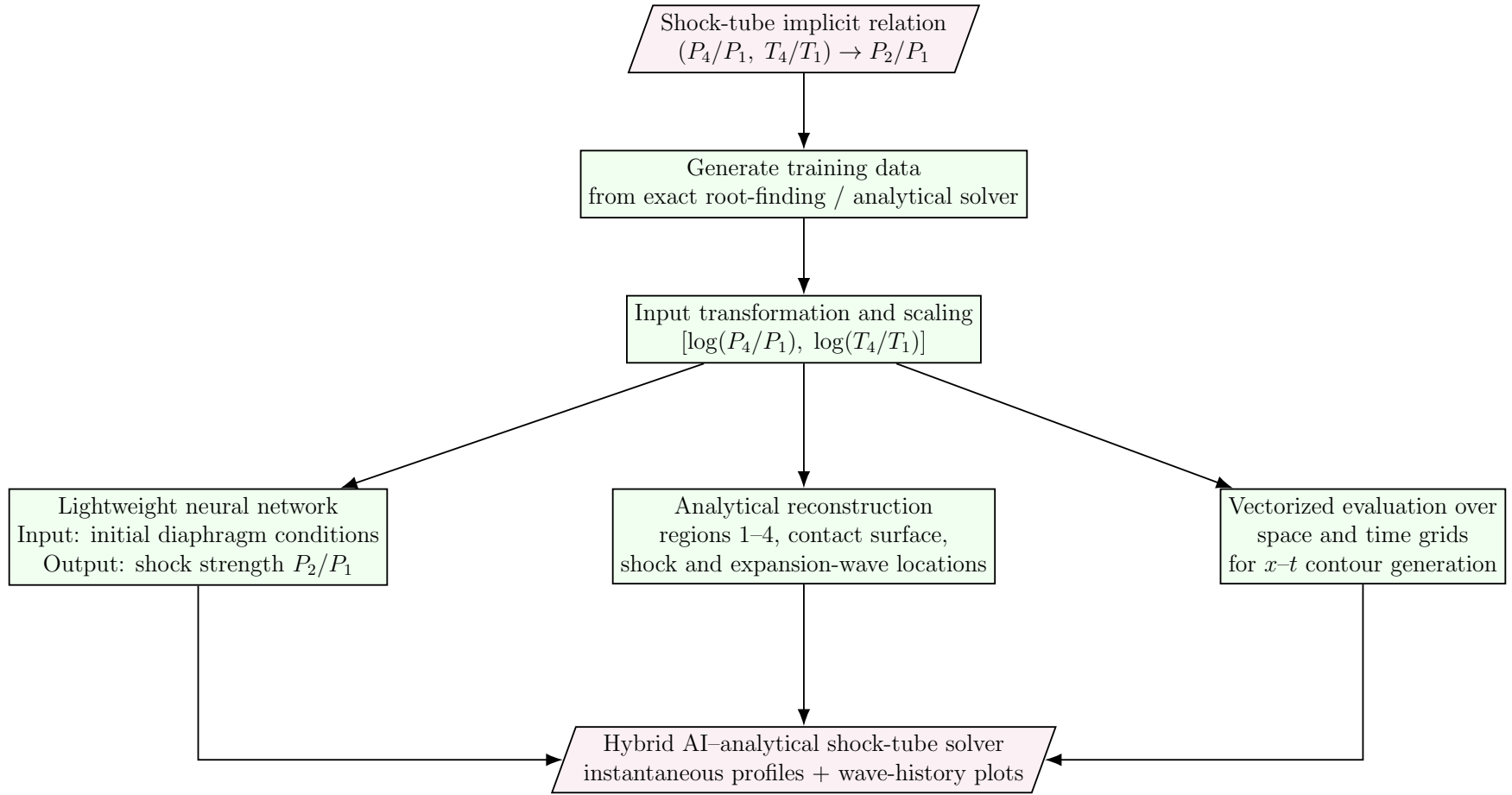


Figure 5: Hybrid AI-analytical workflow for the shock-tube surrogate. Training data are generated from the exact implicit shock-tube relation. The neural network is trained only to predict the shock strength from the initial diaphragm state, while the remaining thermodynamic states and wave trajectories are reconstructed analytically. This hybrid design enables both instantaneous flow-profile prediction and rapid generation of space-time wave diagrams.